

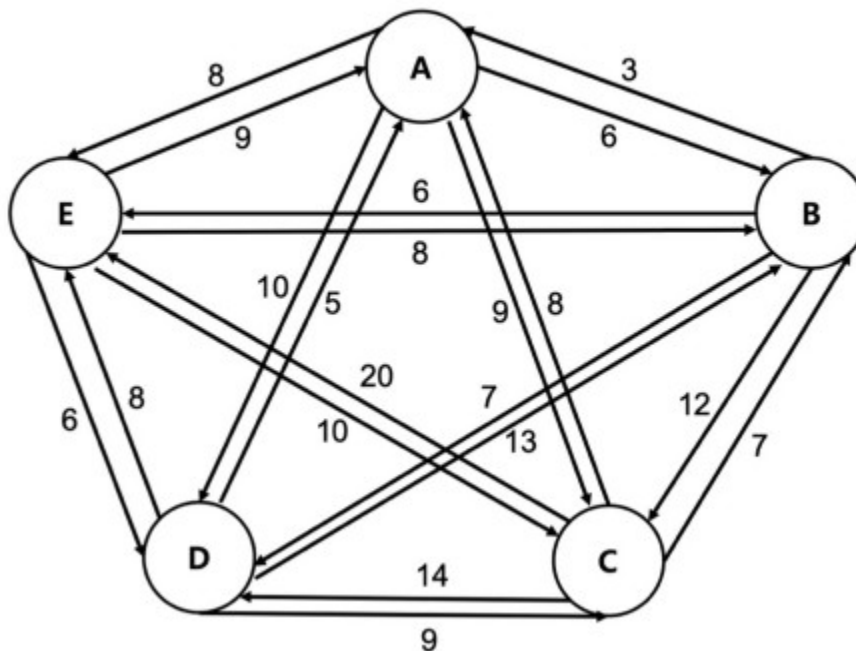
BRAC UNIVERSITY
Department of Computer Science and Engineering
CSE 422: Artificial Intelligence
Assignment -02

Question-01

- A.** What is Elitism in Genetic Algorithm and what could be the impact on the algorithm?
- B.** What is a Single Point Crossover and Two-point crossover? Demonstrate with an example
- C.** What are some stopping conditions Genetic Algo may implement?
- D.** Importance of Mutation in Genetic Algo?
- E.** Importance of Crossover in Genetic Algorithms?

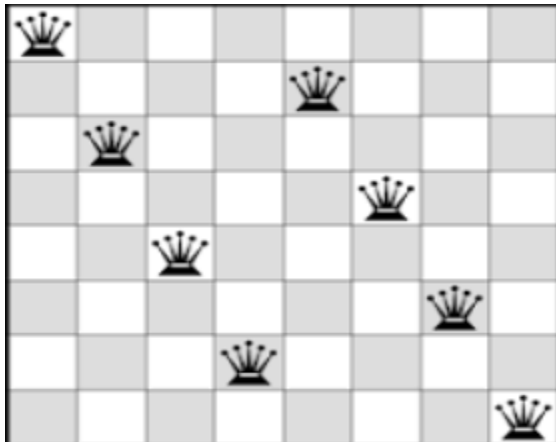
Question-02

The Traveling Salesman Problem (TSP) is a classic optimization problem. Where the goal is to find the shortest possible route that visits each city exactly once and returns to the starting city given a set of cities(a,b,c,d,e) and their path cost.



- a).** Construct a population of 4 chromosomes, define fitness function.
- b).** Select the best two chromosomes using the **Roulette Wheel Selection** or **RankBased** method.
- c).** Now perform **Uniform Cross Over** or **Two-point Crossover** to create a total of 4 children.
- d).** Perform single bit mutation and comment if the children's chromosomes are eligible or not. If not, then does multi-bit mutation solve the problem?

Question-03



Following is 8-Queen board, now simulate Genetic Algorithm for the 8-Queen board.

- Construct a population of 4 chromosomes, define fitness function.
- Select two chromosomes for crossover using natural selection
- Now perform a **Single Point Crossover** to create a total of 4 children.
- Perform mutation
- Calculate their fitness

Question-04

A company uses a machine learning algorithm to recruit its employees. However, there are some complaints that the algorithm has a bias based on certain demographic areas, gender, race etc meaning that two candidates with similar qualities may not be given the same opportunity. To identify if any bias exists the company decides to divide a group of candidates into two subsets such that candidates with similar qualifications go to distinct subsets so they can find out if candidates with similar qualifications are getting different outcomes. The similarity score between 6 candidates (A1... A6) is calculated based on skills, education, experience and is given below:

	A1	A2	A3	A4	A5	A6
A1	0	8	3	6	5	7
A2	8	0	7	4	2	3
A3	3	7	0	9	6	2
A4	6	4	9	0	7	1
A5	5	2	6	7	0	5
A6	7	3	2	1	5	0

The task is to partition the candidates into two subsets such that the similarity scores between candidates in different subsets is maximized and the similarity score between candidates in the same subset is minimized. For example, if two candidates are in two different subsets the fitness should increase, if they are in the same subset then the fitness should decrease.

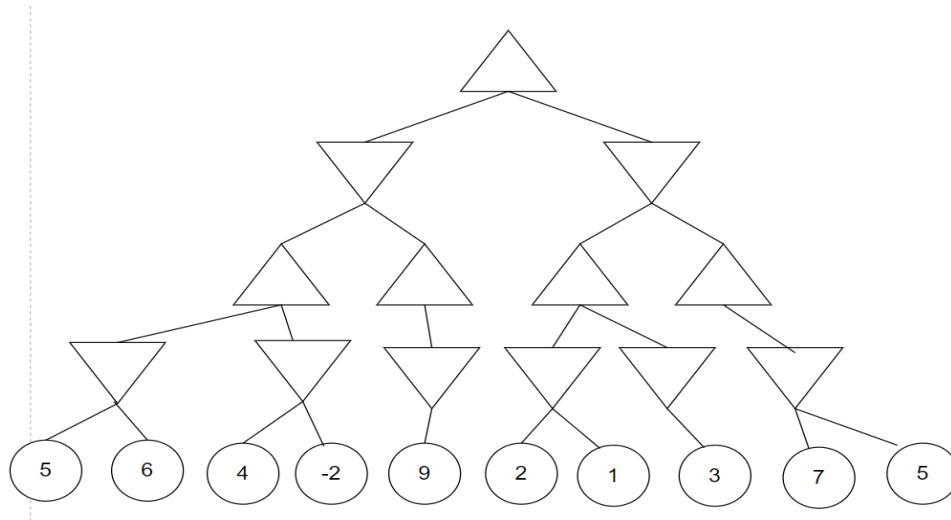
Propose an encoding scheme to represent the solution in a genetic algorithm. Explain using an example.

Define a fitness function to evaluate how good a partition is. Use the example from a. to explain.

Demonstrate the crossover operation between two randomly generated parents. What problem may arise if you run a genetic algorithm with population size two?

Question-05

Perform Alpha-Beta Pruning on the following state space tree, and find the state value, alpha and beta values for all the nodes.



Question-06

1. Is alpha beta pruning a Depth first search or Breadth First Search?
2. Does ordering of the leaf nodes change the outcome of MiniMax Algorithm?
3. Is it possible that MiniMax and Alpha Beta pruning will end up giving the same results?